HUGO ABREU

Senior Developer (Full Stack/Embedded) | AI Specialist | MSc Systems Engineering

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PROFESSIONAL EXPERIENCE

Senior Software Developer

SIDIA

Jun. 2022 – Current Manaus, AM

- PoC development like: Position Estimation of TVs in indoor environment to prevent piracy and Content suggestion, user behavior analysis for VoD via swarm of agencies and content retrieval. Graph network modeling for visually impaired TV navigation via remote control.
- Python/NodeJS/React: Generative AI Animations based on human movements + FastAPI & React. Pose estimation and object detection with mediapipe and yolo respectively. Video processing and similarity detection. NLP (Spacy, NLTK) for PT-BR x LIBRAS dataset generation. Using LSTM (TensorflowJS) and T5 Transformers (HuggingFace) model for Seq2Seq translation. Audio transcription with Whisper and Vosk and native libs in TizenOS. Also creating a full lib in C# from scratch for pt-BR to LIBRAS translation based on semantic and syntax structure changes from both languages.
- Working to deploy the Portuguese-to-LIBRAS translation solution for DTV+ (TV 3.0) broadcasting.
- Developing a NodeJS + React + Electron app (front-end). Using Python for image and video description tasks with CLIP and BLIP models, and applying NLP and Transformer models for paraphrasing (HuggingFace).
- C++ (20, 17, 11), C# (.NET 8): DTVPlus development for Tizen TVs. Maintaining, debugging, and adding new features to C/C++ and C# libraries and applications for Tizen, specifically for cameras and IoT devices.
- Automating firmware performance tests with a C++/Bash app for embedded Linux (TizenOS), using Python Flask for APIs and front-end rendering, Docker + PostgreSQL for the app's database.
- Applying deep learning to verify checkpoint images and automatically classify performance KPIs. Using Python Django RestAPI for data scraping, OpenCV and Scikit-learn + TensorFlow for generating classifications.
- Scientific paper and patents writing.

Prompt Engineer

Outlier

Jun. 2023 – Feb. 2025 Remote | Freelance

Enhancing Large Language Models for specific training needs:

- Reviewing code generated by LLMs in multiple programming languages (Go, Python, C, C++, C#, Java, JavaScript, TypeScript) to address specific prompts.
- Specializing in data science and training large language models to solve diverse prompt-related challenges.
- Working on reinforcement learning from human feedback projects related to prompt generation, code generation, code review, and code fixing for large language models.
- Applying adversarial techniques prompts in chatbots to detect possibility of malware generation.
- Detecting malware generated by LLMs through code reviews and analysis of prompt intent.

SCIENTIFIC PUBLICATIONS

Journal Scientific Papers

- Mendes, H. (2026). 5d spatial brane vs 4d temporal quantum multiverse: A phenomenological comparison of merging-universe models in a modified friedmann framework. *https://doi.org/10.21203/rs.3.rs-10272409/v1*.
- Figueiredo, R., & Mendes, H. (2024). Analyzing information leakage on video object detection datasets by splitting images into clusters with high spatiotemporal correlation. *IEEE Access*, *12*, 47646–47655.
- Mendes, H., & Seixas, P. (2024). Similarity-based ranking of videos from fixed-size one-dimensional video signature. *Discover Computing*, *27*(1), 25.
- Mendes, H. A. [Hugo Abreu]. (2022). Correção de erro de previsão de radiação solar da apac utilizando comitês de perceptrons multicamadas otimizado por algoritmo genético em dois estâgios. *http://revistas.poli.br/index.php/anais/*
- Mendes, H., Nunes, H., & de Mattos Neto, P. (2020). Integração de dados e modelos de previsão de produção fotovoltaica do nordeste brasileiro. *Revista de Engenharia e Pesquisa Aplicada*, *5*(2), 62–72.
- Mendes, H., Nunes, H., Marinho, M., & Neto, P. (2020). Data integration and prediction models of photovoltaic production from brazilian northeastern. *arXiv preprint arXiv:2001.10866*.
- de França, J., Mendes, H., Costa, L., Dantas, A., & Duarte, A. (2019). Using near infrared spectroscopy and machine learning to diagnose systemic sclerosis. *arXiv preprint arXiv:1908.06137*.
- Mendes, H., de França, J., da Costa, M., Dantas, A., & Lins, E. (2017). Protótipo de baixo custo de um sistema imagens da difusão ôptica com radiações no nir para validação de diagn´osticos na pele.

Conference Proceedings

- Cunha Filho, C. A. H., & Mendes, H. A. [Hugo Abreu]. (2024). Tvtat - a real time camera imaging testing tool for smart tvs: Preliminary results. In *Anais do simpósio brasileiro de testes de software sistemático e ...*.
- Costa, L., Campos, T., Maciel, E., Mendes, H., Liaqat, M., & Lins, E. (2022). Um novo tomôgrafo por micro-ondas para imageamento da mama utilizando braços robôticos. In *Congresso brasileiro de automática-cba* (Vol. 3).
- Mendes, H. (2021). On automls for short-term solar radiation forecasting in brazilian northeast. In *International conference on engineering and emerging technologies*.
- Mendes, H. A. [Hugo Abreu], de Oliveira, J. a., Mattos, P., & Coutinho, A. (2021). Metaheuristic based mlp-sarimax hybridization one hour ahead solar radiation forecasting. In *Xv cbic*.

- Generating new prompts from zero to test the capabilities of the LLM and reviewing prompts generated by others to adjust in terms of usability and difficulty levels.
- Reviewing labeling generated by others as a trusted prompt engineer.

Systems Engineer

FITec

Feb. 2019 – May. 2022 Recife, PE

- C# .Net for embedded systems (WindowsCE) and Windows Apps for communication and systems integration, used for supervising and adjusting train routes.
- Development of quality control and automation tools for computer, notebook and all-in-one manufacturers. BASH and Python: FastAPI.
- Deployment and update of Nokia VitalQIP IPAM (IP Address Management) and DNS servers on RHEL servers.
- Automating the massive import of network data into VitalQIP, while maintaining network structure and inputting end-client information.
- Development of automation solutions in Python, Java and BASH, for integration into RHEL servers, communicating with Nokia 77XX routers.
- Development of scripts in Apache Velocity in the Nokia NSP tool to automate the configuration of Nokia 77XX routers.
- Deploying Nokia NSP for router management on RHEL servers.
- Implementation of Nokia AAA (Radius protocol) on CentOS / RHEL server.
- Analyzing and correcting security vulnerabilities in RHEL systems, including hardening of servers.
- Providing solutions for IP networks, including logical upgrades to network infrastructure, and ensuring the consistency of IP pools and network topology.
- Customizing Debian based Linux distribution for OEM installation.
- Multidisciplinary problem solving using machine learning, optimization, statistics and other mathematical and computational tools, such as digital signal and image processing, with Python libraries such as NumPy, Scikit-learn, SciPy, and Pandas.
- Development of web applications (API) and Microservices with computational models using SpringBoot and Flask. Use of the Selenium testing tool for Web Scraping.
- Implementation of a test notebook in the development stage of a smart meter (DLMS Protocol). Automation of the COSEM object structure in XML using Python.

CLP lectures Assistant

CTG UFPE

Aug. 2018 – Dec. 2018 Recife, PE

- Assisting with PLC lectures for Electrical Engineering and Control & Automation courses at UFPE.
- Preparing lectures and solving exercises involving the use and programming of SIEMENS S7-1200 PLCs.
- Industrial instrumentation using stepper motors, AC & DC motors, optical encoders, and variable-frequency drives.

Research Internship

FACEPE

Aug. 2017 – Jul. 2018 Recife, PE

- Developed near-infrared imaging equipment for diagnosing skin lesions.

- Gonçalves, G. L., Klautau J´unior, A. B. d. R., & de, R. (2019). Avaliação de qualidade visual baseada em rede bayesiana aplicada a imagens médicas.
- Mendes, H., Lins, E., de França, J., Dantas, A., & da Costa, M. (2019). Development of a doi equipment using nir radiation for skin lesions diagnosis and blood vessels recognition. In *Xxvi brazilian congress on biomedical engineering: Cbeb 2018, armação de ....*
- Mendes, H., Yaseen, M., Bezerra, M., Nunes, F., Lins, E., & Shaid, S. (2019). Electrochemical impedance spectroscopy applied in theoretical model of exclusion zone. In *Xxvi brazilian congress on biomedical engineering: Cbeb 2018, armação de ....*

Patents

- MENDES, H., FIGUEIREDO, R., CUNHA FILHO, C., & AUZIER BR, B. (2026). Método para homologação de smart tv com aprendizado de contexto personalizado. Patent BR102024016156-4A2.

EDUCATION

Master's Degree in Systems Engineering

University of Pernambuco

March 2020 – January 2022 Recife, BR

Specialization in Artificial Intelligence

UPE

Feb. – Nov. 2019 Recife, BR

Control and Automation Engineering

Federal University of Pernambuco

2013 – 2018 Recife, BR

Electronic Engineering

ENSEEIHT

2015 – 2016 Toulouse, FR

- Involved in mechanical prototyping using SolidWorks, designing photoelectronic circuits, and conducting image and signal processing with Python and MATLAB.
- Performed digital processing of skin infrared light absorbance spectra, utilizing PCA and SVM for diagnosing rare autoimmune diseases.
- Developed an imaging device using infrared LEDs and photosensors. Conducted digital image processing of bovine teeth to study radiation and infrared scattering, and to assess image quality.
- Developed a LabView-based supervisory system for characterizing aqueous samples, communicating via OPC with MATLAB and metrology hardware from National Instruments, MyDAQ.

Internship

WT Engenharia

Oct. 2017 – May. 2018 Recife, PE

- Maintenance and assembly of embedded electronic RF communication module for dew sensing, as a secure protocol, to working with energized high voltage transmission lines.
- IoT project applied to actuation of disconnect switches using microcontrollers (Arduino) and WIFI network (ESP8266), TCP/IP master-slave protocol.

PROGRAMMING & TECH

Python | Pytorch | TensorFlow | LLM / Transformers | SciKitLearn | Pandas | Computer Vision | Time Series | Prompt Engineering | C/C++ & C# | STL | Boost | Glib | GStreamer | Linux & Bash Scripts | JavaScript | React | JAVA

CERTIFICATIONS

Docker: Creating and Managing Containers | Testing in .NET: Creating Unit Tests with xUnit | Developing on AWS (3.4.0) | .NET: Creating a Web API with ASP.NET Core | Hugging Face: Improving Model Efficiency with Optimum

FOREIGN LANGUAGES

Portuguese (native)
English
French
Spanish

